

Brian Sahagian

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EDUCATION

Boston University, College of Engineering

Bachelor of Science in Mechanical Engineering, GPA: 3.94/4.00

Boston, MA

May 2026

PROJECTS

Drone ESC Overheating Root-Cause Analysis, Revolute Robotics

Jun - Jul 2026

- Diagnosed and resolved critical ESC overheating failures on a hybrid mobility robot
- Reduced ESC peak temperatures by 28% via a ventilated enclosure redesign and purpose-built heat sink
- Engineered ESP32 thermal logger and Python analysis pipeline to validate operational temperatures
- Delivered engineering recommendations and drove fleet-wide adoption of hardware solution

Unmanned *E. coli* and Cyanobacteria Monitoring Buoy, Senior Thesis

Sep 2025 - May 2026

- Led mechanical design of real-time bacteria detection device for Boston's Charles River Watershed
- Manufactured weatherproof, watertight, and buoyant housing for electronics and sensors to IP68 rating
- Developed C++ firmware for RF communication across 5 km over a 3-month continuous deployment

Trash Sorting Robotic Arm, Robotic Systems

Mar - May 2026

- Built 4-degree-of-freedom SCARA robotic arm for extracting recyclable materials from mixed waste
- Developed pick-and-place kinematic algorithms using MATLAB Robotics Toolbox and C++ for control
- Manufactured custom arm and end effector with 3D-printed, machined, and laser cut components

Skateboarder Tracking Camera, Harvard University CS50x

Jun - Jul 2026

- Engineered target-tracking camera using Python, OpenCV, and YOLO for self-filming skateboarding
- Built 2-axis camera gimbal using STM32 microcontroller, servo actuators, and 3D-printed components
- Tuned closed-loop PID controllers with anti-windup, deadband, and clamping for stabilized recording

WORK EXPERIENCE

Revolute Robotics

Boston, MA

Robotics Engineering Intern

May 2026 - Present

- Advance mechanical development for a hybrid drive/fly robot for industrial confined-space inspection
- Assemble, repair, and troubleshoot electromechanical systems to maintain operational fleet
- Model design iterations in Fusion, manufacture components in-house for rapid prototype validation
- Execute operational field testing and failure analysis to isolate robot vulnerabilities

Boston University Engineering Product Innovation Center

Boston, MA

Machine Shop Assistant

Dec 2024 - May 2026

- Supervised machine shop floor, assisting and teaching students proper use of machining equipment
- Optimized manufacturing processes using custom test fixtures and CNC programming
- Enforced strict safety and maintenance protocols, ensuring secure shop environment for students

SKILLS

Programming: MATLAB, C, C++, Python, ROS2, SQL, Git

Design & Simulation: SolidWorks, Onshape, Fusion, MATLAB Robotics Toolbox, CAM

Mechanical: Manual & CNC machining, mill, lathe, 3D printing (FDM, SLA), laser cutter, waterjet, soldering

Technical: Mechanical drawings, precision assembly, Engineering Validation Testing, DMM, stress testing, oscilloscope, GD&T, Root Cause Corrective Action, SOPs, sub-assemblies, AI-assisted development

ACTIVITIES

- Classic motorcycle restoration, specializing in rebuilding and maintenance of air-cooled 1970's Hondas
- Boston University Snowboarding Team, competing in slalom and freestyle events across the Northeast
- Tau Beta Pi Honor Society, facilitating service events to foster engineering networking across campus